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1 June 1971
SUPERSEDING
PA-PD-2789
23 December 1965

MILITARY SPECIFICATION

# CARTRIDGE, SEMI-FIXED, 105MM, APERS-T XM546E2, LOADING, ASSEMBLING, AND PACKING

## 1. SCOPE

1.1 This specification covers the loading, assembling and packing for the Cartridge, 105MM, Anti-Personnel, XM546E2 for Howitzer M2A2, M103 and M137.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-A-2550 | - Ammunition and Special Weapons, General Specification for |
| MIL-I-45607 | - Inspection Equipment, Supply and Maintenance of |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105) |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-414 | - Sampling Procedures and Tables for Inspection by Variables for Percent Defective |
| MIL-STD-1168 | - Lot Numbering of Ammunition |
| MIL-STD-1235 | - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |

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DRAWINGS

ARMY

| | | |
|---|---|---|
| 8862347 | - | Box, Packing, Ammunition, for 105MM Howitzer Ammunition in Fiber Container M472 |
| 8862348 | - | Container, Ammunition Fiber M472 for 105MM Howitzer Ammunition |
| 9241622 | - | Projectile, 105MM: Anti-Personnel, XM380E6 Loading Assembly |
| 9241623 | - | Cartridge, Semi-Fixed, 105MM Anti-Personnel XM546E2 for Howitzer Cannon M2A2, M103, M137 Assembly |

PUBLICATIONS

ARMY

ET-9241623 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Cartridges.-The cartridges shall comply with all requirements specified on Drawing (dwg.) 9241623 and with all requirements specified in applicable specifications.

3.3 Primer.-The primer shall have the required number and location of flash holes as specified on the appropriate primer drawing.

3.4 Proving ground.-Testing shall be accomplished as specified in 4.3.2.

3.4.1 Cartridge functioning.-The cartridges shall function satisfactorily as evidenced by first and second projectile function.

3.4.1.1 Time-to-burst.-When set for 5.0 seconds, first projectile function shall occur after 4.7 seconds and prior to 5.6 seconds.

3.4.1.2 Muzzle action.-When set for Muzzle Action (MA), first projectile function shall occur prior to 5 meters from the muzzle and second projectile function shall occur prior to 55 meters from the point of first function.



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3.4.2 Muzzle Velocity.-The muzzle velocity of the cartridge, when tested at 70 degrees Fahrenheit (°F), shall not be more than 1830 feet per second (fps) nor less than 1770 fps and the standard deviation shall not exceed 8.0 fps.

3.4.3 Indicator function.-The tracer shall exhibit a satisfactory function by emitting a visually observable incendiary flame. The dye charge shall exhibit a satisfactory function by producing a visually observable puff of smoke indicative of first function.

3.4.4 Early burst.-The round shall not function before 75 percent of set time when set for 5.0 seconds (i.e. 3.75 seconds).

3.4.5 Security of parts.-There shall be no loss of components or metal parts separation in the gun bore, the muzzle brake, or in flight, prior to fuze function.

3.5 Workmanship

3.5.1 All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. All parts shall be free of chips, cracks, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.5.2 The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such painting and markings have been marred or obliterated.

3.5.3 Rotating bands.-Care shall be exercised at all times so that the rotating band of the projectile is not nicked, burred or otherwise damaged.

3.5.4 Adhesive on threads.-Threads shall be clean and free of contaminants prior to the application of the specified quantity of adhesive (see 4.2.3.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.



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4.1.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.2 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information, accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions.

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn, and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105.



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Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity critiera specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded cartridges shall contain:

a. Cartridge cases of one type of metal from one manufacturer in accordance with the same drawing and drawing revision, same specification and specification revision.
b. Propelling charges from not more than one lot.
c. Loaded primers of one lot interfix number from one manufacturer.
d. Projectiles of one lot interfix number from one manufacturer.
e. Loaded relays of one lot interfix number from one manufacturer.
f. Loaded detonators of one interfix number from one manufacturer.
g. Tracers of one lot interfix number from one manufacturer.
h. Fuzes of one interfix number from one manufacturer.
i. Smoke marker pellets of one interfix number from one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.



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4.2.2.1 Relay and detonator assembly (see dwg. 9230474 covering a detail of dwg. 9241622).

| Categories | Defects | Method of Inspection | Code No. (see 6.2) |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | 100 Percent Inspection | | |
| 101. | Detonator or relay improperly assembled or missing | Visual | 01001 |
| Major: | AQL 0.40 percent | | |
| 102. | Contents loose | Visual | 01002 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship, (see 3.5) | Visual | 01003 |

4.2.2.2 Fuze adapter and smoke marker loading assembly (see dwg. 9211472 covering a detail of dwg. 9241622).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Any component missing or improperly assembled | Visual | 02001 |
| Minor: | AQL 1.50 percent | | |
| 201. | Distance from flash tube to outside face | Gage | 02002 |
| 202. | Total length | Gage | 02003 |
| 203. | Evidence of poor workmanship, (see 3.5) | Visual | 02004 |

4.2.2.3 Base charge assembly (see dwg. 9211477 covering a detail of dwg. 9241622).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Weight of propellant | Scale | 03001 |
| 102. | Bag torn or heat seal open | Visual | 03002 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship, (see 3.5) | Visual | 03003 |

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4.2.2.4 Projectile and fuze adapter loading assembly prior to assembling (see dwg. 9241622).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | 100 Percent Inspection | | |
| 101. | Relay and detonator housing assembly missing | Visual | 04001 |
| 102. | Radial detonator missing or improperly assembled | Visual | 04002 |
| 103. | Adhesive on threads inadequate (see 3.5.4 and 4.2.3.5) | Visual | 04003 |
| Minor: None defined. | | | |

4.2.2.5 Projectile after loading base charge cavity (see dwg. 9241622).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | 100 Percent Inspection | | |
| 101. | Base charge loading assembly missing (prior to covering) | Visual | 05001 |
| Major: | AQL 0.40 percent | | |
| 102. | Propellant cover missing | Visual | 05002 |
| Minor: None defined. | | | |

4.2.2.6 Projectile prior to assembling (see dwg. 9241622).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | 100 Percent Inspection | | |
| 101. | Adhesive on threads inadequate (3 places) (see 3.5.4 and 4.2.3.5) | Visual | 06001 |
| Major: | AQL 0.40 percent | | |
| 102. | Tracer missing or inverted (prior to assembling plug and disc assembly) | Visual | 06002 |



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| | | | |
|---|---|---|---|
| Minor: | AQL 1.50 percent | | |
| 201. | Marking missing, misleading or unidentifiable | Visual | 06003 |
| 202. | Protective coating with total damaged area in excess of 1/4 square inch | Visual | 06004 |
| 203. | Evidence of poor workmanship, (see 3.5) | Visual | 06005 |

4.2.2.7 Loaded primer, prior to assembly to cartridge case (see dwg. 8866664 covering a detail of dwg. 9241623).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Any flash hole missing | Visual | 07001 |
| Major: None defined. | | | |
| Minor: None defined. | | | |

4.2.2.8 Cartridge case prior to assembling propelling charge (see dwg. 9241623).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Primer above flush | Gage | 08001 |
| Major: None defined. | | | |
| Minor: None defined. | | | |

4.2.2.9 Cartridge (see dwg. 9241623).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: None defined. | | | |
| Minor: | AQL 1.00 percent | | |
| 201. | Marking missing, misleading or unidentifiable | Visual | 09001 |
| 202. | Evidence of poor workmanship, (see 3.5) | Visual | 09002 |



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4.2.2.10 Unsealed fiber container (see dwg. 8862348)

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Any packing component missing or improperly assembled | Visual | 10001 |
| Minor: | AQL 1.50 percent | | |
| 201. | Projectile and case cannot be removed by hand | Manual | 10002 |
| 202. | Container caps cannot be removed by hand | Manual | 10003 |
| 203. | Glue or asphalt on cartridge | Visual | 10004 |

4.2.2.11 Sealed fiber container (see dwg. 8862348).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Tape incomplete or badly wrinkled | Visual | 11001 |
| 102. | Container cut or damaged thru all asphalt impregnated layers | Visual/ Manual | 11002 |
| 103. | Barrier bag torn or missing | Visual | 11003 |
| Minor: | AQL 2.50 percent | | |
| 201. | Tear tab length, minimum | Gage | 11004 |
| 202. | Gap between cover and body container exceeding 1/8 inch | Visual | 11005 |
| 203. | Marking missing, misleading or unidentifiable | Visual | 11006 |
| 204. | Cuts, scuffs, or gouges in outer layers | Visual | 11007 |
| 205. | Contents loose | Manual | 11008 |



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4.2.2.12 Sealed wood packing box (mee dwg. 8862347).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Protective bag missing (prior to closing) | Visual | 12001 |
| 102. | Box damaged exposing contents | Visual | 12002 |
| 103. | Hardware or strapping missing, broken or loose | Visual | 12003 |
| 104. | DOD symbol missing or incorrect | Visual | 12004 |
| Minor: | AQL 1.50 percent | | |
| 201. | Strapping or hardware improperly assembled | Visual/ Manual | 12005 |
| 202. | Handle missing or insecure | Visual/ Manual | 12006 |
| 203. | Car seal missing, unsealed or improperly positioned | Visual | 12007 |
| 204. | Marking missing, misleading or unidentifiable | Visual | 12008 |

4.2.3 Testing

4.2.3.1 Bourrelet Diameter (see dwg. 9241622) - Major defect - code No. 06003.-The projectile loading assembly shall be gaged 100 percent for maximum diameter forward and rearward of the rotating band. Any projectile which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.2.4.

4.2.3.2 Torque (see dwg. 9241622).

| | | |
|---|---|---|
| Projectile Base | Major | 06004 |
| Fuze adapter | Major | 06005 |
| Fuze | Major | 06006 |
| Plug and disc assembly | Major | 06007 |

After the cement has dried, twenty projectiles shall be randomly selected from each lot for this test. If any projectile fails to withstand the torque when tested in accordance with 4.3.1, the lot shall be rejected.



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4.2.3.3 Check test for possible deterioration (see applicable primer and detonator loading specifications) - Major defect - Code No. 13001.-If the total elapsed time between the original acceptance of any primer or detonator lot and the assembly of that lot into cartridges exceeds two years, or if the primer or detonator have been subject to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to, and must satisfactorily pass the check test for deterioration specified in the applicable loading specification immediately before the primers or detonators are assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (see 6.4).

4.2.3.4 Mouth diameter and maximum profile of the cartridge case (see applicable cartridge case drawing) - Major defect - Code No. 13002.-The cartridge cases shall be gaged 100 percent for profile and diameter of the mouth. Any case which fails to comply with the applicable requirements shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.2.4. (NOTE: This inspection shall not be required if it is determined that the cartridge cases have been previously gaged 100 percent for these characterisitcs.

4.2.3.5 Adhesive on threads (see dwg. 9241622).-The adequacy of the manufacturing application method shall be quantitatively assessed on one sample for each threaded joint at the start of each production shift, as well as each time there is a change in the method or process that is likely to affect uniformity of performance as determined by the Government. If the sample fails to meet the drawing requirement, production shall be stopped and the application method corrected. When it has been determined that the quantity applied by the application method in use meets the drawing requirement, visual observation shall be used to assure that this amount is applied to each projectile assembly (see 4.2.2.4 and 4.2.2.6).



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4.2.3.6 Proving Ground (see 3.4).-Defects shall be classified in accordance with the following table: The tests shall be performed as specified in 4.3.2.

| Categories | Defects | Code No. |
|---|---|---|
| Critical | Break-up prior to clearing gun bore or muzzle brake | 14001 |
| Major | Detonation of the round prior to 3.75 seconds when set for 5.0 seconds | 14002 |
| Major | Cartridge fails to function satisfactorily when set for 5.0 seconds as evidenced by first projectile function at or prior to 4.7 seconds or at or after 5.6 seconds | 14003 |
| Major | Cartridge fails to function satisfactorily when set for MA as evidenced by first projectile function at or after 5 meters | 14004 |
| Major | Cartridge fails to function satisfactorily when set for MA as evidenced by second projectile function at or after 55 meters from first function | 14005 |
| Major | Muzzle velocity exceeds 1830 fps or is less than 1770 fps | 14006 |
| Major | Standard deviation of muzzle velocity exceeds 8.0 fps | 14007 |
| Major | Tracer fails to emit a visually observable incendiary flame | 14008 |
| Major | Dye charge fails to produce a visually observable puff of smoke indicative of first function | 14009 |

4.2.3.6.1 Time-to-burst (see 3.4.1.1) - Code No. 14003

4.2.3.6.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with all the requirements of 3.4.1.1, 3.4.3 and 3.4.4, 25 sample cartridges shall be randomly selected from each lot for this test. The sampling plan and procedures for this test shall be in accordance with Standard MIL-STD-414, Section B, Part II, Table B-3, Code Letter I, and an AQL of 2.5 percent. In addition, if three or more malfunctions (i.e. duds, unsatisfactory functions or outliers - see 6.7) occur, the lot shall be rejected.



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4.2.3.6.1.2 After three consecutive lots have complied with 4.2.3.6.1.1, 10 sample cartridges shall be randomly selected from each lot for this test. The sampling plan and procedures for this test shall be in accordance with Standard MIL-STD-414, Section B, Part II, Table B-4, Code Letter I, and an AQL of 2.5 percent. In addition, if two or more malfunctions (i.e. duds, unsatisfactory functions or outliers - see 6.7) occur, the lot shall be rejected. In the event that three successive lots fail to meet all the criteria of this paragraph the sampling provisions of 4.2.3.6.1.1 shall again apply until three successive lots meet all the requirements, contained therein.

4.2.3.6.2 Early burst (see 3.4.4) - Code No. 14002.-All cartridges fired for observation of time-to-burst shall also be observed for early bursts. If any early burst occurs, the lot shall be rejected.

4.2.3.6.3 Muzzle velocity (see 3.4.2) - Code No's 14006 and 14007.-All cartridges fired for observation of time-to-burst shall also be observed for muzzle velocity. The sampling plan and procedures for this test shall be in accordance with MIL-STD-414, as referenced in 4.2.3.6.1.1 or 4.2.3.6.1.2 as applicable, except that a 1.5 percent AQL shall be used. For the first three lots produced, 10 cartridges from the reference calibration lot used for propellant assessment shall be fired alternately with the test rounds and appropriate velocity correction factors shall be applied. For subsequent lots, calibration rounds will not be fired with the original test. If the maximum allowable percent defective specified in MIL-STD-414 is exceeded for the subsequent lots, the lot shall be referred. Also, if there is more than one outlier (see 6.7) or lost value, the lot shall be referred. In addition, if the standard deviation of the sample cartridge muzzle velocities multiplied by the factor 0.75 when the sample size is 25 rounds or 0.65 when the sample size is 10 rounds exceeds 8.0 fps, the lot shall be rejected. In the event that the lot is referred due to muzzle velocity, an additional 10 test cartridges shall be randomly selected and fired alternately with 10 cartridges from the reference calibration lot available for propellant assessment. The sampling plan shall be in accordance with MIL-STD-414, as referenced in 4.2.3.6.1.2. Identical instrumentation and the same weapon components used for the first test shall be used for this second test.

4.2.3.6.4 Muzzle action.-First projectile function (see 3.4.1.2) - Code No. 14004.-In addition to the cartridges fired for 4.2.3.6.1.1 or 4.2.3.6.1.2, 10 additional sample cartridges shall be randomly selected from each lot for this test. The sampling plan and procedures for this test shall be in accordance with Standard MIL-STD-414, Section B, Part I, Form 2, Table B-3, Code Letter F, and an AQL of 1.5 percent. In addition, if two or more malfunctions (i.e. duds, unsatisfactory functions or outliers-see 6.7) occur, the lot shall be rejected.

4.2.3.6.5 Muzzle Action - Second projectile function (see 3.4.1.2) - Code No. 14005.-All cartridges fired for observation of first projectile function shall also be observed for second function. If two or more sample cartridges fail to meet the second projectile function requirement, the lot shall be rejected.



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4.2.3.6.6 Indicator function (see 3.4.3) - Code No.'s 14008 and 14009.-All cartridges fired for observation of time-to-burst and muzzle action functioning shall also be observed for indicator function. If three or more sample cartridges fail to comply with the tracer requirement or if four or more sample cartridges fail to comply with the dye charge requirement when 35 cartridges are fired then the lot shall be rejected. If two or more sample cartridges fail to comply with the tracer requirement or if three or more sample cartridges fail to comply with the dye charge requirement when 20 samples are fired, then the lot shall be rejected.

4.2.3.6.7 Security of parts (see 3.4.5) - Code No. 14001.-All cartridges fired for these tests shall be observed for security of parts. If any sample cartridge fails to meet the security requirement, the lot shall be rejected.

4.2.3.6.8 Test validity.-If for any reason the proving ground considers that the test conditions detrimentally affected the test results, the test shall be considered invalid and a new test shall be performed with additional samples.

4.2.4 Inspection Equipment.-Index of Inspection Equipment Lists Number ET-9241623 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Torque test.-The specified torque shall be applied in the disassembly direction. A thin line shall be drawn across the joint in question to aid in observing movement. The projectile shall be secured properly to avoid interference of other joints not under test.



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4.3.2 Proving ground tests.-This test shall be performed at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired at charge zone 7 in a weapon system using the M103 series or M137 series tube. Appropriate radar coverage, with necessary auxiliary equipment such a reflector screen, shall be used in conjunction with photographic coverage and visual observation to determine compliance with the applicable requirements. Projectile muzzle velocity shall be recorded for the time-to-burst phase only. Each projectile shall be visually observed for satisfactory function of its tracer and dye charge and for information only, the smoke marker pellet. In addition, chamber pressure shall be recorded for information on each round tested. Means and standard deviations should be recorded for all data whenever possible.

4.3.2.1 Muzzle Action Phase.-Adequate photographic coverage, including smear cameras, shall be provided to determine compliance with the first and second function distance requirements and to witness the metal parts integrity up to and including second function with any metal parts failures after fuze function recorded for information. In addition, for information only, two (2) samples shall be observed for total target hits over two foot intervals and the width of the hit pattern on a target 80 ft wide by 4 ft high at a distance of 100 meters and the data recorded. The aiming point shall be at the lateral center of the target with an elevation of 35 mils above the vertical center of the target.

4.3.2.2 Time-to-burst phase.-The sample cartridges for this phase shall be temperature conditioned to 70°F plus or minus 2 degrees for at least 16 hours. No more than five minutes shall elapse from the time the round is removed from the conditioning chamber until the time it is fired. In addition, the sample cartridges fired for this phase shall be fired with the fuzes set for 5.0 seconds and observed for compliance with the time-to-burst requirement as evidenced by first projectile function. Muzzle velocity shall be recorded for each sample cartridge fired for time-to-burst to determine compliance with the requirement. At the end of this phase remaining tube life shall be determined and recorded on the firing record for informational purposes.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with dwg. 8862348.

5.2 Packing

5.2.1 Level A.-The containers shall be packed in accordance with dwg. 8862347.

5.3 Marking.-Marking shall be in accordance with dwgs. 8862347 and 8862348.



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## 6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification

b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167.

6.2 Inspection code Numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment
MU - Army Design, mandatory for use



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6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.



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6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND12. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal Completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Cost of check tests.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the primer or detonator or the contractor loading and assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., while the primers or detonators were under the jurisdiction of either contractor.

6.5 It is noted that the size of inspection lots of components or assemblies may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests and deliveries, inspection lots may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.6 Submission of ballistic test data and ammunition data cards.-In addition to the normal distribution of records for loaded items provided by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to Commanding Officer, Picatinny Arsenal, Dover, New Jersey, 07801, ATTN: SMUPA-ND2 and SMUPA-DC10.

6.7 Outlier criteria.-All the values of the sample shall be used to calculate the sample mean and standard deviation. The standard deviation value obtained shall then be used in the determination of outliers (non-normal values) by using the criteria of the mean plus and minus 3 standard deviations. A value judged to be an outlier shall be excluded and the remaining values in the sample used to recalculate the mean and standard deviation.



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6.8 Proving ground test summary

| Test | Sample Size (See 4.2.3.6.1 & 4.2.3.6.4) | | Requirements |
|---|---|---|---|
| Time-to-burst | #*25 | #10 | a. Functioning (see 3.4.1.1)<br>b. Indicator function (see 3.4.3)<br>c. Early burst (see 3.4.4)<br>d. Security (see 3.4.5)<br>#e. Velocity (see 3.4.2)<br>f. Pressure for information (see 4.3.2) |
| MA Phase | *10 | 10 | a. Functioning (see 3.4.1.2)<br>b. Indicator function (see 3.4.3)<br>c. Security of parts (see 3.4.5)<br>d. Metal parts integrity for information (see 4.3.2.1)<br>e. Hit dispersion for information 2 samples (see 4.3.2)<br>f. Pressure for information (see 4.3.2) |
| Total sample cartridge required | *35 | 20 | |

*Applicable for first three lots or tightened inspection

#In the event that uncorrected muzzle velocity fails to meet the requirements of 3.4.2, the lot shall be referred, and an additional 10 test cartridges shall be fired (see 4.2.3.6.3). Corrected muzzle velocities are required for the first three lots produced.

Custodian:

ARMY -MU

Preparing Activity:

ARMY-MU



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-50252 (MU)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE ARMY

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DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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